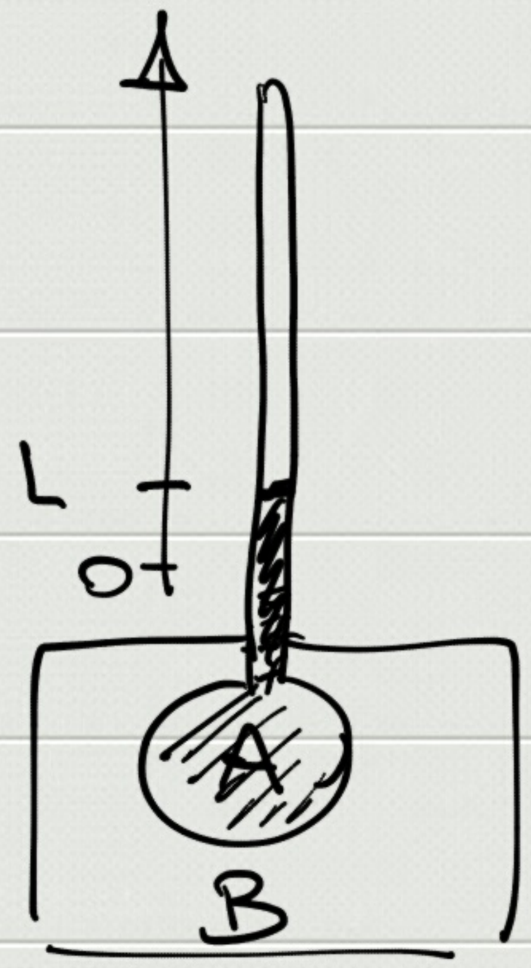
equilibrio termico $\Rightarrow$ temperatura



termometro : sistema TD

$$x = x(t)$$

temperatura

caratteristica termometrica

funzione termometrica

$$\boxed{t = \theta(x)}$$

temperatura

$\theta \rightarrow$ funzione monotona (crescente)

$$t = \theta(L) = aL + b$$

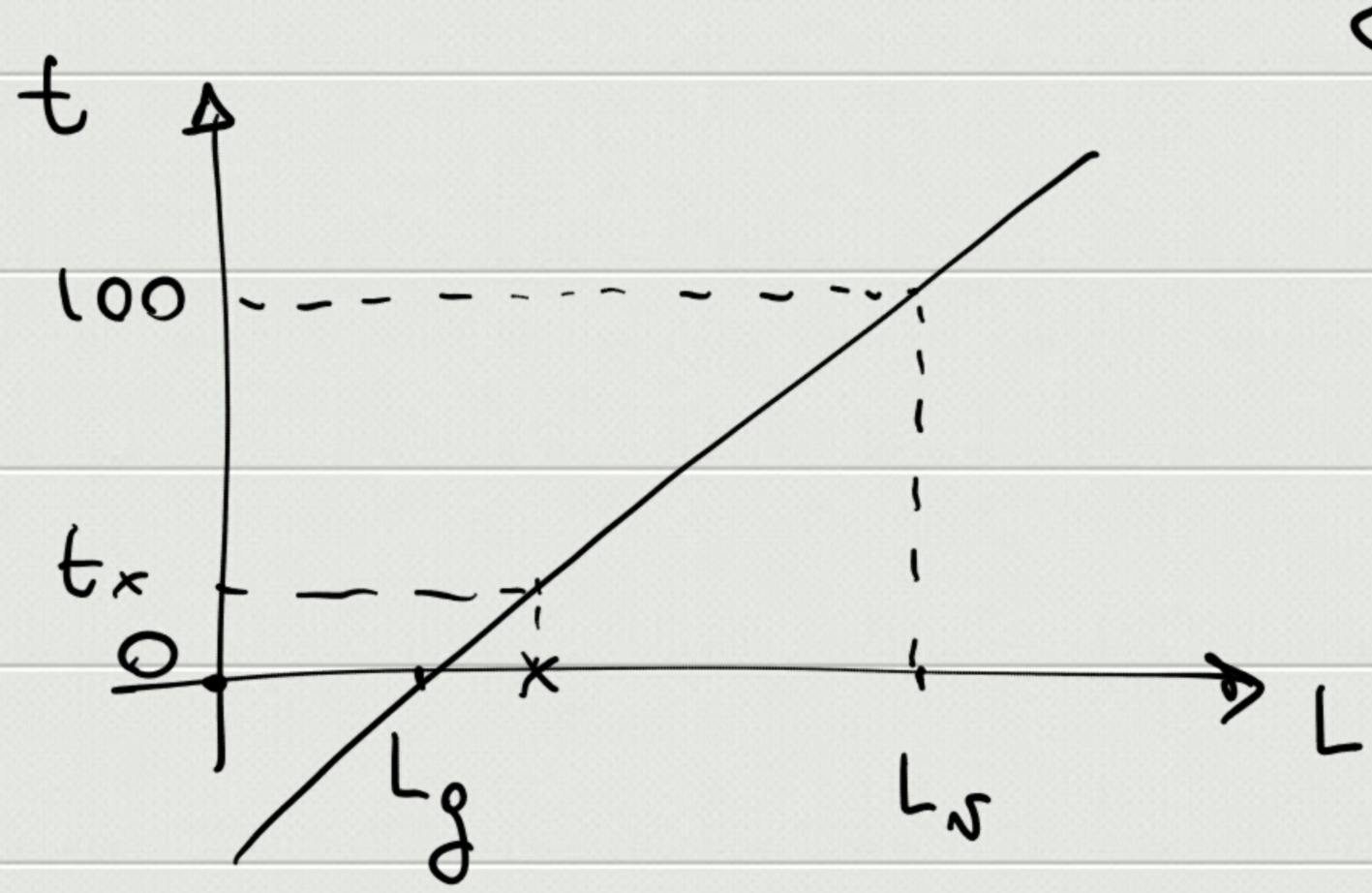
fusione ghiaccio : $t = 0 = a L_g + b$

ebollizione acqua : $t = 100 = a L_v + b$

$$100 = a(L_v - L_g) \Rightarrow$$

$$a = \frac{100}{L_v - L_g} \qquad b = -aL_g = -100\,\frac{L_g}{L_v - L_g}$$

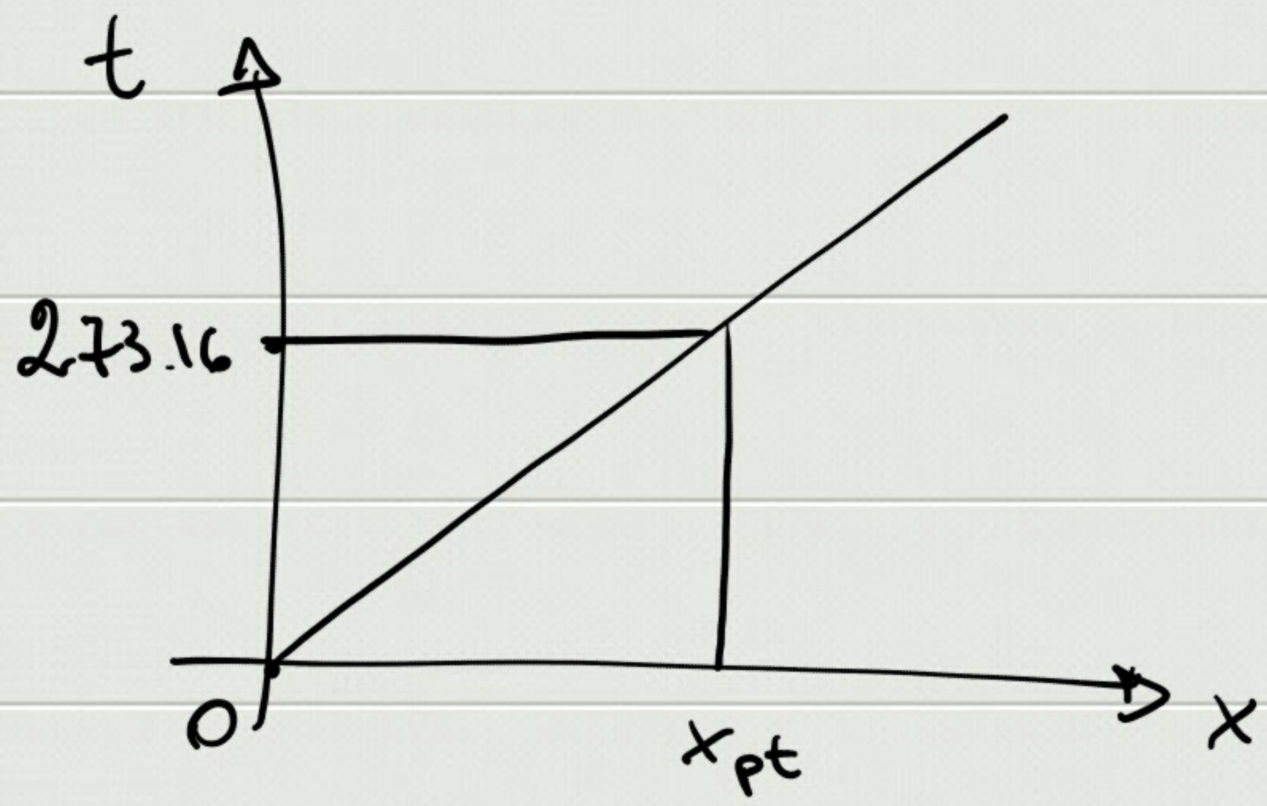
$$\Rightarrow t = \theta(L) = aL + b = 100\,\frac{L - L_g}{L_v - L_g}$$



Scala centigrada : Celsius (°C)

$\Theta(x) = a\,x$ $\qquad$ $t_{pt} = \Theta(x_{pt}) = a\,x_{pt} = 273.16$

$$\Rightarrow a = \frac{273.16}{x_{pt}} \Rightarrow \boxed{t = \Theta(x) = 273.16\,\frac{x}{x_{pt}}}$$



temperature empiriche

$\Rightarrow T_{min} \equiv \emptyset \Rightarrow$ temperatura assoluta $\underline{\underline{T}}$

unità di misura è il <u>Kelvin</u> (K)

$T_g = 273.15$ K

$T_{pt} = 273.16$ K $\qquad$ $\Delta T\,(K) = \Delta t\,(°C)$

$T_v = 373.15$ K